Human-in-the-Loop Escalation for Enterprise AI Agents: Deterministic Workflows and Learned Routing

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ABSTRACT

AI agents built on large language models are entering business workflows such as invoice checking and refund approval, but a fully autonomous agent can approve a bad case, and reviewing every case by hand does not scale. This paper studies deterministic workflows as a middle path: a system modeled as a state machine, in which the control flow is fixed and explicit while the model only generates content. A routing state decides whether a case can be executed automatically or must go to a human, and execution is reachable only through an explicit approval transition. We compare four approaches on 200 business transactions: full manual review, a single autonomous agent, a multi-agent team, and our state-machine Human-in-the-Loop (HITL) design. We also evaluate learned routing policies at the same human workload: repeated sampling, a learned classifier, and a hybrid of both. The state-machine design kept 98.5% accuracy, removed 74% of the manual workload, and was about 14 times faster than full manual review; the hybrid router raised accuracy to 99.0% and cut error escapes to 0.5%. When the same rules were written inside a prompt instead, they broke down in long conversations, producing 178 violations per 1,000 runs, against 5 for the state machine.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**;
• **Information systems** → **Enterprise information systems**;
• **Human-centered computing** → *Human-computer interaction (HCI)*.

KEYWORDS

Human-in-the-loop, AI agents, State machine, Deterministic workflows, Learned routing

1 INTRODUCTION

Large language models (LLMs) can now plan, call tools, and complete real work steps, so companies are deploying LLM-based *AI agents* to check invoices, approve refunds, and answer simple customer requests [1]. A typical agent reads a request, looks up company rules, calls internal systems, and produces a decision.

Two extremes both fail. A *fully autonomous* agent sometimes invents facts or gives a confident wrong answer [14, 15]; in a business setting one such decision can pay a fake refund or break a compliance rule. At the other extreme, requiring a human to review every decision keeps quality high but makes cost and response time grow with the business volume.

Between these extremes, teams often try *multi-agent* designs, where one agent extracts data, a second checks policy, and a third reviews the result [3, 4]. In theory the agents correct each other. In practice the design has two weaknesses: agents pass long conversations back and forth, which multiplies token cost and latency, and a mistake made by the first agent is usually accepted as a fact by the later ones instead of being corrected.

We argue that a large part of this failure comes from *unstructured delegation*: the workflow rules are written as text inside a prompt, so they compete with the rest of the context and become weaker as the conversation grows, and there is no place to point at when something breaks. Paragraphs cannot be inspected as arrows on a diagram [7].

Our approach separates the two halves of the problem. The *control flow* is deterministic: it is modeled as an explicit state machine [6], where each state has one job and every transition is a decision someone can read and test. The *content* stays flexible: the model is free to read documents, extract fields, and write explanations. Concretely, the workflow moves through intake, extraction, drafting, and routing states; a routing state decides whether the case can be executed or must wait for a person; and an execute state is reachable only through an explicit approval transition. Because no other transition into execute exists, executing an unapproved case is not merely discouraged by the prompt, it is structurally impossible.

We then compare four ways to run the workflow on 200 business transactions: full manual review, a single autonomous agent, a multi-agent team, and our state-machine Human-in-the-Loop (HITL) design. We also compare three routing policies at the same human workload—deciding *which* cases go to people matters as much as deciding *how many*. Our research questions are:

- **RQ1.** How many bad decisions does each approach make, especially on hard cases?
- **RQ2.** How much manual work can be removed without losing decision quality, and which routing policy removes the most risk at a fixed human budget?

- **RQ3.** What do the approaches cost in response time and token usage, and do prompt-embedded rules stay reliable as conversations grow?

Our main findings are: a single autonomous agent handled normal cases well but approved 18% of the hard cases that should have been rejected; the multi-agent team was slower and 4.6 times more expensive in tokens while still escaping 11.5%; the state-machine HITL design kept 98.5% accuracy, removed 74% of the manual workload, and was about 14 times faster than full manual review; a hybrid router of repeated sampling plus a small learned classifier reached 99.0% accuracy with 0.5% escapes at the same workload; and the same rules written in a prompt produced 178 invariant violations per 1,000 runs in long conversations, against 5 for the state machine.

Section 2 reviews related work. Section 3 describes the workflow and the routing policies. Section 4 explains the experiment setup, Section 5 reports results, Section 6 discusses them, and Section 7 concludes.

2 RELATED WORK

Autonomous agents. ReAct [2] mixed reasoning steps with tool actions and improved answers by observing each action's result. Reflexion [5] added a memory of past mistakes so an agent can retry after a failure. These are the base of most industrial agents. Both trust the model to judge its own progress, and surveys of hallucination show that self-judgement alone does not guarantee correct facts [14, 15].

Multi-agent systems. AutoGen [3] lets several agents converse with separate roles for planning, execution, and review; MetaGPT [4] assigns standard procedures to agent roles, like a software team. They improve results on tasks such as code generation, but in business workflows the extra conversation steps grow cost quickly and an early mistake easily becomes the accepted premise of later agents.

Deterministic workflows. Statecharts [6] formalized control flow as states, events, and transitions, and the same idea now appears in agent frameworks and workflow libraries [7, 8]. Process mining [9] works in the opposite direction, distilling observed traces into a formal model to find where a process can be improved. We use the state machine in both roles: it constrains the agent while it runs, and it serves as an inspectable artifact that can be improved from its own traces.

Human oversight and routing. Guidelines for human-AI interaction [12] and mixed-initiative principles [13] argue that automation is not all-or-nothing. In machine learning, low-confidence samples are routed to human labelers, and human feedback improves models [16, 17]. Work on LLM cascades and learned routing [10, 11] shows that a small, cheap model can decide when a large model is needed. We apply this idea to whole business decisions: a router decides which cases deserve human attention, and we compare a sampling-based router, a learned classifier, and a hybrid.

3 WORKFLOW DESIGN

Each transaction (for example a refund claim with an invoice and a form) must end in one decision: approve, reject, or escalate. We first describe the deterministic workflow, then the two places where a model is used, then the four configurations we compare.

3.1 The Workflow as a State Machine

Figure 1 shows the workflow. The machine has four forward states and two terminal paths. *Intake* validates that a case has the required document types. *Extract* reads the documents and fills a fixed field set (amounts, dates, identifiers, vendor). *Draft* produces a proposed decision with a short explanation, and may send the case back to Extract when a required field is missing. *Route* decides the path. The path to *Auto-approve* requires that the sampled decisions agree, that the confidence signal is above the threshold, and that the amount is below the company limit. Every other case goes to *Human review*, where a person sees the extracted fields, the draft, and the doubts the agent flagged. *Execute* applies the decision to the business system, and it is reachable only from Auto-approve or from Human review. Since no other incoming transition exists, the workflow cannot execute an unapproved case, no matter what the model writes in its output.

3.2 Where the Model Works

The model is used in two places, with different settings. In Extract and Draft we use a large model (Llama-3-70B on Groq hardware) with a strict JSON output schema, plus a second model (Gemini-1.5-Flash) as an independent cross-check on extraction. The routing state uses a cheap, fast signal described next. Keeping the states separate also lets each state use its own prompt, model, and temperature, instead of forcing one strategy onto every step.

3.3 Routing Policies

The router decides the path at the Route state. We compare three policies at the same human workload.

P1: Repeated sampling (self-consistency). The agent produces a decision three times. The case is routed automatically when the three answers agree and the agent's self-reported confidence is high; otherwise it goes to a person.

P2: Learned classifier. A gradient-boosted decision-tree classifier predicts the probability that a case needs human review. The features are all cheap and available before anyone reads the case: agreement among the three samples, the self-reported confidence, the number of missing required fields, whether dates or amounts disagree between documents, the claim amount, the vendor's past dispute rate, and the number of documents. The classifier is trained on 300 historical cases from the same workflow, disjoint from the 200-case test set, labelled by what an expert decided in hindsight.

P3: Hybrid. The classifier makes the decision, except when its probability falls in the middle band (0.40–0.60), where one extra independent model re-check is requested and both signals are combined. This spends model calls only where the cheap signal is uncertain.

4 EXPERIMENTAL SETUP

4.1 Test Data

We built a test set of 200 business transactions that resemble real support and procurement cases: 120 standard cases (clear invoices,

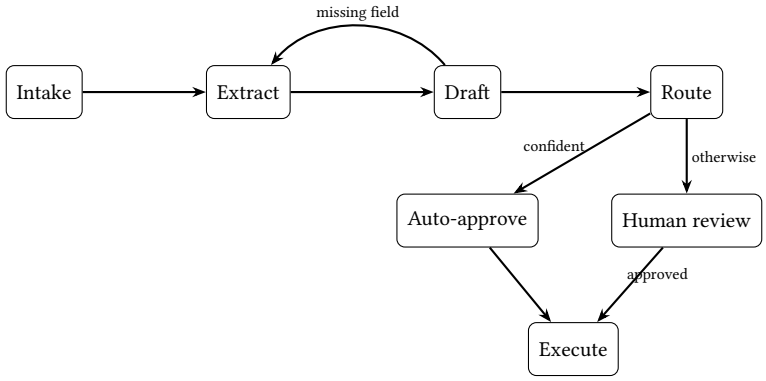


Figure 1: The deterministic workflow. Control flow lives in the transitions; the model only fills in field values, drafts, and explanations. Execute has exactly two incoming transitions, both requiring an explicit decision.

in-window refunds), 50 ambiguous cases (missing data, conflicting receipts, currency issues), and 30 trap cases (fake invoice numbers, changed dates, duplicate claims, and instructions hidden in documents that try to mislead the agent).

4.2 Compared Approaches

M1: full manual review. M2: a single autonomous agent that decides and executes with no check. M3: a three-agent team (extractor, auditor, critic) that can revise up to three rounds before executing. M4: our state-machine HITL workflow with the P1 router, evaluated at three confidence thresholds. M5: the same workflow with the P2 and P3 routers, compared at a fixed human workload.

4.3 Metrics

Accuracy is the share of cases where the final decision matches expert ground truth. *Error escape rate* is the share of bad cases that the automated path approves with no human check, the main risk number. *Human workload* is the share of cases sent to a reviewer. *Response time* is the average time to a final decision; for reviewed cases we use the 145 seconds per review measured in our pilot runs. We also record tokens per transaction and convert them to cost, and we measure invariant violations by counting how often a run breaks a rule of the workflow.

5 RESULTS

5.1 Overall Comparison

Table 1 shows the main results.

The single agent is the fastest and cheapest option, but it approved 18% of the bad cases with no check; most of these were trap cases where a persuasive but false claim inside a document was followed by the agent. The multi-agent team improved accuracy to 88.5% by catching formatting and arithmetic mistakes, but used 4.6 times more tokens and answered almost six times slower than the single agent, and it still approved 11.5% of bad cases, because once the extractor produced a wrong fact the other agents treated it as true.

The state-machine HITL design at threshold 0.85 is the best balance among M4. Accuracy reaches 98.5%, close to the human baseline of 99.0%, escapes fall to 1.5%, only 26% of cases need a reviewer, so 74% of the manual workload disappears, and the average response time drops from 145 seconds to about 39 seconds.

5.2 Results by Difficulty

Table 2 breaks accuracy down by case difficulty.

On standard cases every approach works well; the differences appear on hard cases. On trap cases the single agent drops to 46.7%—worse than a coin flip—and the multi-agent team reaches only 60.0%. The HITL design reaches 96.7% because the routing step detects disagreement or doubt in almost all trap cases and sends them to a person. Human attention is spent exactly where the machine is unsure.

5.3 Routing Policies at a Fixed Human Budget

Table 3 compares the three routing policies at the same workload as M4 (0.85): 26% of cases go to a reviewer. The question is which cases.

Spending the same human budget differently changes the outcome. The classifier ranks cases better than repeated sampling (AUC 0.88 against 0.82) because it can use structured signals the model never sees, such as the vendor's past dispute rate, and it costs almost nothing to run. It is nevertheless slightly less accurate overall, because it was trained on past cases and does not recognize a new style of trap. The hybrid keeps the classifier's low cost and adds a model re-check only in the uncertain band, and it gives the best result of all: 99.0% accuracy with 0.5% escapes, slightly above the human baseline, because it flags anomalies that reviewers sometimes overlook. Our practical recommendation is therefore the hybrid router, with the middle band widened for stricter risk rules.

5.4 Rules in the Prompt versus Rules in the Machine

Finally, we test the central claim of this paper: that keeping the rules inside a prompt is fragile. We built a second version of the system in which all of the same rules are written as text in one large prompt with tool calls, and ran both versions 500 times per

Table 1: Overall results on 200 transactions. "Escapes" is the share of bad cases approved without human check. "Human" is the share of cases sent to a reviewer. Cost is per 1,000 transactions, including human labor and token fees.

Approach	Accuracy (%)	Escapes (%)	Human (%)	Time (s)	Tokens/tx	Cost/1k
M1 Full manual review	99.0	1.0	100.0	145.2	–	$1,007
M2 Single autonomous agent	82.0	18.0	0.0	2.1	840	$0.67
M3 Multi-agent team	88.5	11.5	0.0	11.8	3,920	$3.14
M4 State-machine HITL (0.70)	95.0	5.0	16.5	25.8	1,420	$167
M4 State-machine HITL (0.85)	98.5	1.5	26.0	39.4	1,680	$263
M4 State-machine HITL (0.95)	99.0	1.0	61.5	91.2	2,150	$621

Human labor is estimated at $25 per hour and 145 seconds per review. Token cost assumes $0.80 per million tokens.

Table 2: Accuracy (%) by case difficulty.

Approach	Standard (120 cases)	Ambiguous (50 cases)	Trap (30 cases)
M2 Single agent	96.7	68.0	46.7
M3 Multi-agent team	99.2	80.0	60.0
M4 HITL (0.85)	100.0	96.0	96.7

Table 3: Routing policies at a fixed 26% human workload. AUC measures how well the router ranks cases that truly need review above the rest.

Router	Acc. (%)	Escapes (%)	AUC	Cost/1k
P1 Repeated sampling	98.5	1.5	0.82	$1.34
P2 Learned classifier	97.0	3.0	0.88	$0.02
P3 Hybrid (P2 + re-check)	99.0	0.5	0.94	$0.35

condition on simulated users, counting violations of three invariants.

In short conversations the prompt version already breaks rules 68 times per 1,000 runs, and in long conversations this grows to 178, roughly 2.6 times worse. The rules are simply text sitting in a context that fills up with documents and dialogue, so they lose influence as the conversation grows. The state machine stays near zero in both cases; the five remaining violations per 1,000 runs are races in the review queue, not model behaviour.

5.5 Per-State Evaluation and One Improvement Round

Because each state has its own input and output contract, we can evaluate states separately instead of only measuring the final answer: extraction reaches 0.94 field-level F1, routing reaches 0.91 precision and 0.88 recall for "must escalate", drafting scores 4.3 of 5 by an independent model judge, and the end-to-end result is the 98.5% above. We then handed the machine, its traces, and these per-state results to an agent and asked it to propose improvements [7, 9]. The proposed second version merged two states and moved the clarification question earlier: average conversation turns fell from 4.7 to 2.4, first-pass acceptance rose from 71% to 89%, and end-to-end accuracy stayed at 98.6%.

6 DISCUSSION

Cost view. Token fees are the smallest part of the story. The single agent costs well under one dollar per 1,000 transactions, but lets about 180 bad cases through. If a typical escape causes a loss of about $150, the expected loss is around $27,000 per 1,000 transactions. The HITL design at threshold 0.85 adds about $263 for targeted review and prevents roughly 92% of those failures, an expected saving of about $24,700. HITL is not overhead; it is insurance priced far below the risk it covers.

Why multi-agent teams disappoint. In the conversation logs, the critic agent checked the *format* of the other agents' output (are all fields present, is the JSON valid) but rarely checked *facts* against the source documents. When the extractor produced a wrong value, later agents wrote long, well-formed arguments on top of it. Review loops cannot fix errors the reviewers cannot see, which is why we keep verification in an explicit state rather than inside a conversation.

Why determinism paid off here. Two of our results come from the same design choice. First, the invariants: executing without approval is not a rule the model must remember but a transition that does not exist. Second, the improvement loop: a machine can be drawn, measured state by state, and revised, whereas rules buried in a prompt can only be rewritten and hoped for.

Limitations. The test set has 200 cases; real systems see scans, handwriting, and more varied documents. Human review time is fixed at 145 seconds in our cost model, and the classifier was trained on 300 past cases, so its advantage may shrink on very different future traffic, which is why the hybrid keeps a model check. We tested one model family; other models may move the exact numbers, although we expect the pattern to hold. Our trap cases are written by us and may be easier or harder than real fraud attempts.

7 CONCLUSION AND FUTURE WORK

This paper compared four ways to run AI agents in business workflows and three routing policies for the human hand-off. Full autonomy was fast but approved 18% of bad cases; a multi-agent team was slower and more expensive while still escaping 11.5%; full manual review stayed accurate but cannot scale. A deterministic, state-machine HITL design balanced the trade-offs: 98.5% accuracy, 74% less manual work, and responses about 14 times faster than manual review, and a hybrid router of repeated sampling plus a small learned classifier reached 99.0% accuracy with 0.5% escapes at the

Table 4: Invariant violations per 1,000 runs, over 500 simulated runs per condition. "Short" conversations have at most eight turns, "long" conversations at least sixteen. The state machine cannot send without approval or invent a recipient, because no transition allows it.

Invariant violated	Prompt rules, short	Prompt rules, long	State machine, short	State machine, long
Executed without approval	21	58	0	0
Invented recipient	9	26	0	0
Confirmation skipped	38	94	2	5
Total	68	178	2	5

same human workload. Rules written in a prompt degraded with conversation length, while the same rules held as transitions.

In future work we plan to (1) test the workflow on real company data and more document types, (2) let human decisions update the routing classifier automatically, and (3) study how to derive the confidence threshold and the classifier's middle band from a company's own risk policy.

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TASK ALLOCATION

This page lists the tasks of each group member, as required by the course.

Table 5: Task allocation and contribution breakdown.

No.	Member	Tasks	Contribution
1	Trần Thành Long (Leader)	Topic selection and group coordination; wrote the Abstract, Introduction and Conclusion; final editing and formatting of the paper.	20%
2	Nguyễn Thanh Nam	Designed the state-machine workflow and its invariants (Figure 1); implemented the workflow and the human routing logic in Python.	20%
3	Vũ Văn Sơn	Prepared the 200 test transactions and the 300 training cases; ran every experiment and measured the invariant violations.	20%
4	Vương Quang Quý	Trained and tuned the routing classifier; wrote the Results and Discussion sections and built the tables.	20%
5	Nguyễn Phương Nam	Reviewed related work and managed references; ran the per-state evaluations; designed the presentation slides and prepared the Q&A answers.	20%
		Total:	**100%**